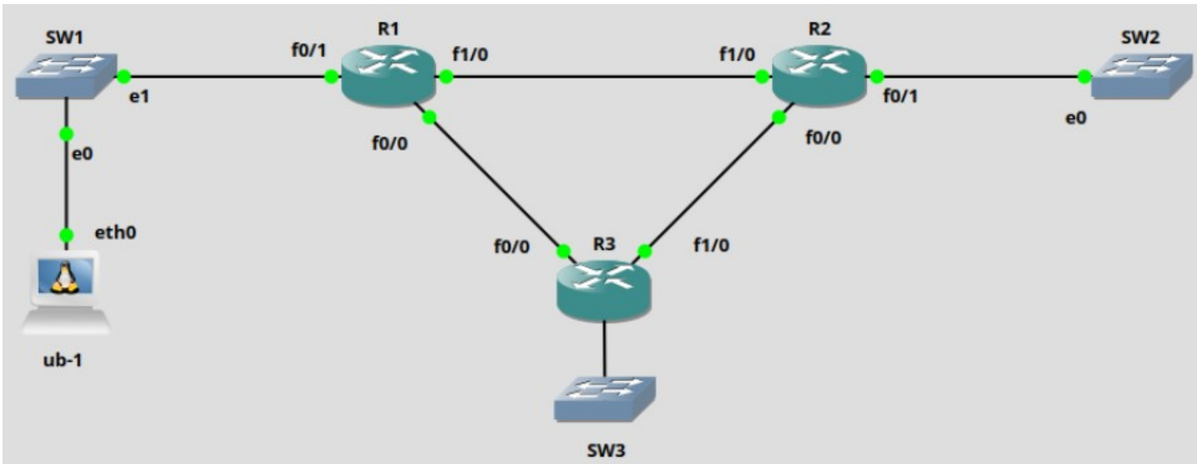
Guião 7 – Routing

Simão Tavares nº 131330



Configuração do OSPF no R1:

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#netw
R1(config-router)#network 10.0.0.0 0.255.255.255 area 0
R1(config-router)#network 192.168.0.0 0.0.255.255. area 0
                                                  ^
% Invalid input detected at '^' marker.

R1(config-router)#network 192.168.0.0 0.0.255.255 area 0
```

Configuração do OSPF no R2:

```
FastEthernet1/0            10.1.2.2        YES manual up
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#netw
R2(config-router)#network 10.0.0.0 0.255.255.255 area 0
R2(config-router)#net
R2(config-router)#network 192.168.0
*Mar  1 00:12:39.431: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.
NG to FULL, Loading Done
R2(config-router)#network 192.168.0.0 0.0.255.255 area 0
```

Configuração do OSPF no R3:

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#
*Mar  1 00:12:26.455: %SYS-5-CONFIG_I: Configured from consol
R3(config)#router ospf 1
R3(config-router)#network 10.0.0.0 0.255.255.255 area 0
R3(config-router)#net
R3(config-router)#network 192.168.0.0 0.0.255.255 area 0
```

Tabela de encaminhamento do R1:

```
R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
C       10.1.3.0 is directly connected, FastEthernet0/0
C       10.1.2.0 is directly connected, FastEthernet1/0
O       10.2.3.0 [110/11] via 10.1.3.2, 00:00:56, FastEthernet0/0
                 [110/11] via 10.1.2.2, 00:00:56, FastEthernet1/0
C    192.168.1.0/24 is directly connected, FastEthernet0/1
O    192.168.2.0/24 [110/11] via 10.1.2.2, 00:00:56, FastEthernet1/0
O    192.168.3.0/24 [110/20] via 10.1.3.2, 00:00:56, FastEthernet0/0
```

Tabela de encaminhamento do R2:

```
R2#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
O       10.1.3.0 [110/11] via 10.1.2.1, 00:01:29, FastEthernet1/0
C       10.1.2.0 is directly connected, FastEthernet1/0
C       10.2.3.0 is directly connected, FastEthernet0/0
O    192.168.1.0/24 [110/11] via 10.1.2.1, 00:01:29, FastEthernet1/0
C    192.168.2.0/24 is directly connected, FastEthernet0/1
O    192.168.3.0/24 [110/20] via 10.2.3.2, 00:01:29, FastEthernet0/0
```

Tabela de encaminhamento do R3:

```
R3#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
C       10.1.3.0 is directly connected, FastEthernet0/0
O       10.1.2.0 [110/2] via 10.2.3.1, 00:01:59, FastEthernet1/0
C       10.2.3.0 is directly connected, FastEthernet1/0
O    192.168.1.0/24 [110/12] via 10.2.3.1, 00:01:59, FastEthernet1/0
O    192.168.2.0/24 [110/11] via 10.2.3.1, 00:01:59, FastEthernet1/0
C    192.168.3.0/24 is directly connected, FastEthernet0/1
```

As routing tables de cada router apresentam as redes ligadas diretamente ao router e as redes aprendidas por OSPF, redes estas que foram anunciadas pelos routers que estão ligados a essa rede

3.

Custo das interfaces de R1:

FastEthernet0/1 is up, line protocol is up
  Internet Address 192.168.1.1/24, Area 0
  Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 10

FastEthernet1/0 is up, line protocol is up
  Internet Address 10.1.2.1/30, Area 0
Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1

FastEthernet0/0 is up, line protocol is up
  Internet Address 10.1.3.1/30, Area 0
Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 10

Custo das interfaces de R2:
FastEthernet0/1 is up, line protocol is up
 Internet Address 192.168.2.2/24, Area 0
Process ID 1, Router ID 192.168.2.2, Network Type BROADCAST, Cost: 10

FastEthernet1/0 is up, line protocol is up
 Internet Address 10.1.2.2/30, Area 0
Process ID 1, Router ID 192.168.2.2, Network Type BROADCAST, Cost: 1

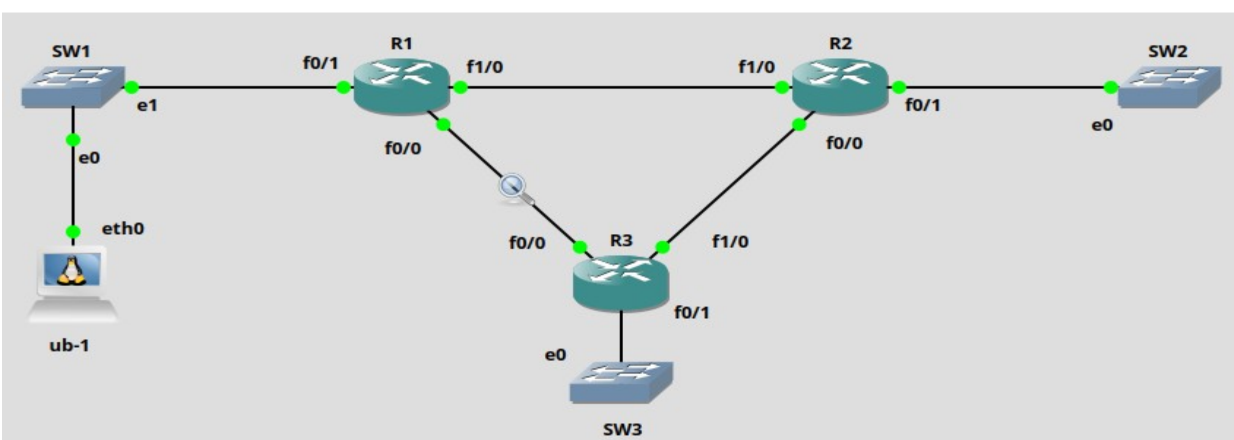
FastEthernet0/0 is up, line protocol is up
 Internet Address 10.2.3.1/30, Area 0
Process ID 1, Router ID 192.168.2.2, Network Type BROADCAST, Cost: 10

Custo das interfaces de R3:
FastEthernet0/1 is up, line protocol is up
 Internet Address 192.168.3.3/24, Area 0
Process ID 1, Router ID 192.168.3.3, Network Type BROADCAST, Cost: 10

FastEthernet1/0 is up, line protocol is up
 Internet Address 10.2.3.2/30, Area 0
Process ID 1, Router ID 192.168.3.3, Network Type BROADCAST, Cost: 1

FastEthernet0/0 is up, line protocol is up
 Internet Address 10.1.3.2/30, Area 0
Process ID 1, Router ID 192.168.3.3, Network Type BROADCAST, Cost: 10

4.



```
R1#conf t
Enter configuration commands,
R1(config)#int f1/0
R1(config-if)#ip os
R1(config-if)#ip ospf co
R1(config-if)#ip ospf cost 20
R1(config-if)#end
```

```
root@ub-1:~# traceroute 192.168.2.2
traceroute to 192.168.2.2 (192.168.2.2), 30 hops max, 60 b
 1  192.168.1.1 (192.168.1.1)  8.657 ms  18.877 ms *
 2  10.1.3.2 (10.1.3.2)  39.197 ms  49.227 ms  59.383 ms
 3  10.2.3.1 (10.2.3.1)  69.585 ms  79.614 ms  89.807 ms
```

*- [R1 FastEthernet0/0 to R3 FastEthernet0/0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

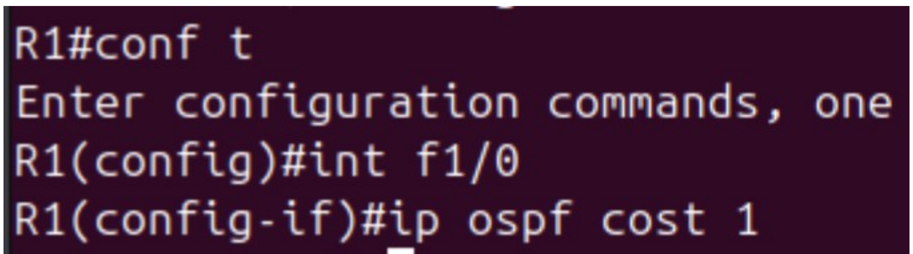
No.	Time	Source	Destination	Protocol	Length	Info
1…	240.74…	192.168.1.…	192.168.2.2	ICMP	98	Echo (ping) request id=0x00
1…	241.74…	192.168.1.…	192.168.2.2	ICMP	98	Echo (ping) request id=0x00
1…	242.74…	192.168.1.…	192.168.2.2	ICMP	98	Echo (ping) request id=0x00

```
R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
C       10.1.3.0 is directly connected, FastEthernet0/0
C       10.1.2.0 is directly connected, FastEthernet1/0
O       10.2.3.0 [110/11] via 10.1.3.2, 00:31:38, FastEthernet0/0
C    192.168.1.0/24 is directly connected, FastEthernet0/1
O    192.168.2.0/24 [110/21] via 10.1.3.2, 00:31:38, FastEthernet0/0
O    192.168.3.0/24 [110/20] via 10.1.3.2, 00:31:38, FastEthernet0/0
```

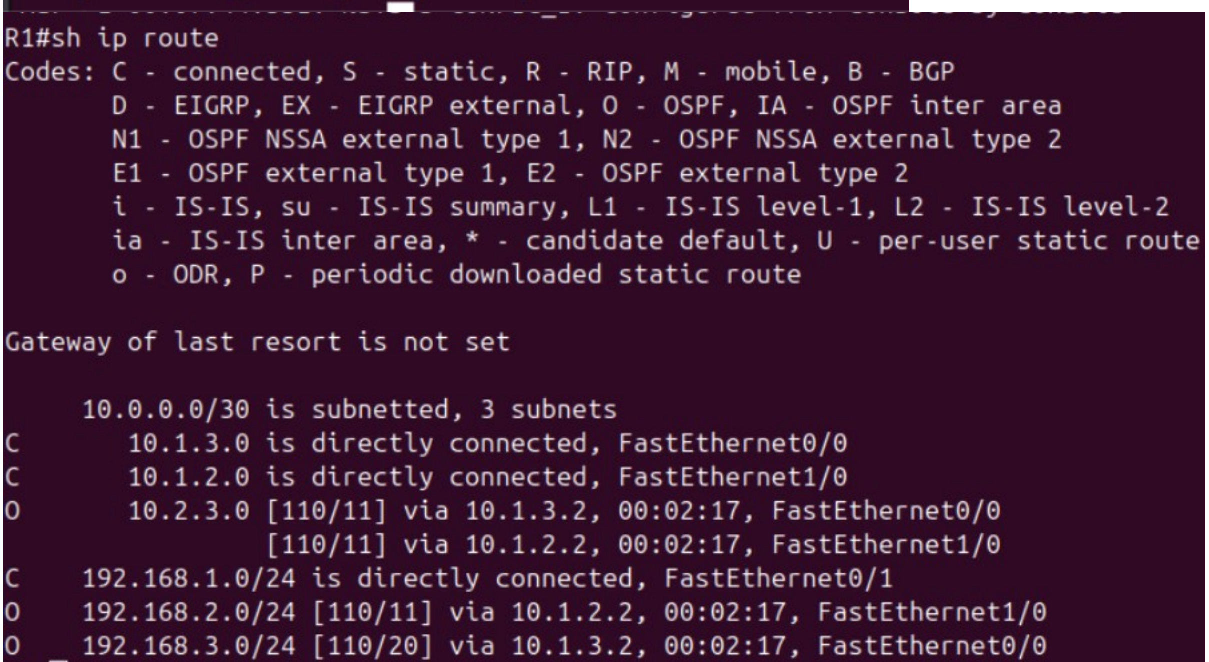
Definimos a porta f1/0 com um custo 20 e mantemos a porta 0/0 com custo 10, assim, o OSPF vai utilizador o caminho com menor custo e os pacotes ICMP vão ser encaminhados pela rota que passa no R3 e, em seguida, no R2. Por fim, teria que se atribuir um custo maior à porta f0/0 do R2, mas neste caso, essa porta já tem custo maior que a porta 1/0, então os pacotes já irão pelo caminho de cima

Verificamos na routing table do R1, que o next-hop para a rede 192.168.2.0/24 mudou devido ao aumento do custo da porta 1/0. Além disso, o custo da rota também aumentou

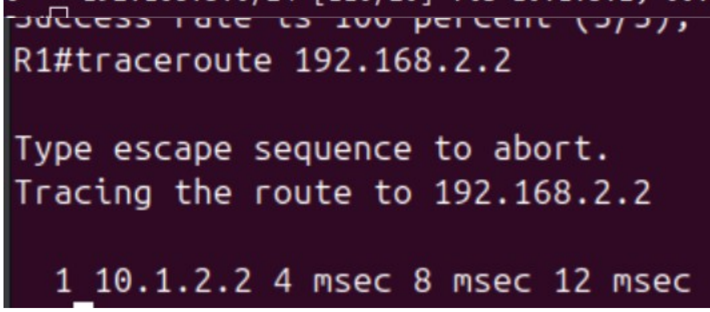
5.



```
R1#conf t
Enter configuration commands, one
R1(config)#int f1/0
R1(config-if)#ip ospf cost 1
```



```
R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
C       10.1.3.0 is directly connected, FastEthernet0/0
C       10.1.2.0 is directly connected, FastEthernet1/0
O       10.2.3.0 [110/11] via 10.1.3.2, 00:02:17, FastEthernet0/0
                 [110/11] via 10.1.2.2, 00:02:17, FastEthernet1/0
C    192.168.1.0/24 is directly connected, FastEthernet0/1
O    192.168.2.0/24 [110/11] via 10.1.2.2, 00:02:17, FastEthernet1/0
O    192.168.3.0/24 [110/20] via 10.1.3.2, 00:02:17, FastEthernet0/0
```



```
R1#traceroute 192.168.2.2

Type escape sequence to abort.
Tracing the route to 192.168.2.2

  1 10.1.2.2 4 msec 8 msec 12 msec
```

Alterámos o custo da porta 1/0 do R1 para 1, ficando com o mesmo custo da porta 1/0 do R2. Com isto, os pacotes já não irão em sentido anti-horário, como no caso anterior, pois o caminho entre o R1 e o R2 passou a ser o caminho com menor custo. Verificámos, na tabela de encaminhamento, a mudança na rota usada, em que agora o next-hop já não é a porta 0/0 do R3, mas sim a porta 1/0 do R2.

6.

```
R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 2 subnets
C       10.1.3.0 is directly connected, FastEthernet0/0
C       10.1.2.0 is directly connected, FastEthernet1/0
C    192.168.1.0/24 is directly connected, FastEthernet0/1
```

```
R2#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 2 subnets
C       10.1.2.0 is directly connected, FastEthernet1/0
C       10.2.3.0 is directly connected, FastEthernet0/0
C    192.168.2.0/24 is directly connected, FastEthernet0/1
```

```
R3#sh ip rou
R3#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 2 subnets
C       10.1.3.0 is directly connected, FastEthernet0/0
C       10.2.3.0 is directly connected, FastEthernet1/0
C    192.168.3.0/24 is directly connected, FastEthernet0/1
```

Ao desligarmos o OSPF, as rotas aprendidas por OSPF são removidas da routing table dos 3 routers.

R1(config)#int f0/0
R1(config-if)#ip ospf 1 area 0

R1(config-if)#int f1/0

R1(config-if)#ip ospf 1 area 0

R1(config-if)#int f0/1

R1(config-if)#ip ospf 1 area 0

R2(config)#int f0/0

R2(config-if)#ip ospf 1 area 0

R2(config-if)#int f0/1

R2(config-if)#ip ospf 1 area 0

R2(config-if)#int f1/0

R2(config-if)#ip ospf 1 area 0

R3(config)#int f0/0

R3(config-if)#ip ospf 1 area 0

R3(config-if)#int f0/1

R3(config-if)#ip ospf 1 area 0

R3(config-if)#int f1/0

R3(config-if)#ip ospf 1 area 0

R1:

```
Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
C       10.1.3.0 is directly connected, FastEthernet0/0
C       10.1.2.0 is directly connected, FastEthernet1/0
O       10.2.3.0 [110/11] via 10.1.3.2, 00:00:33, FastEthernet0/0
                 [110/11] via 10.1.2.2, 00:00:33, FastEthernet1/0
C    192.168.1.0/24 is directly connected, FastEthernet0/1
O    192.168.2.0/24 [110/11] via 10.1.2.2, 00:00:33, FastEthernet1/0
O    192.168.3.0/24 [110/20] via 10.1.3.2, 00:00:33, FastEthernet0/0
```

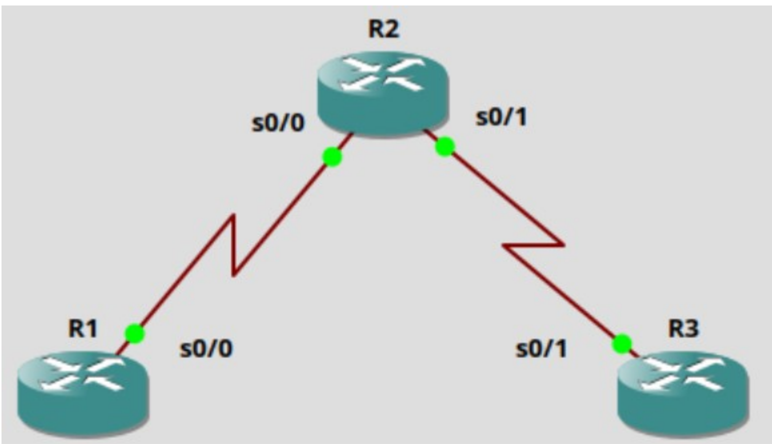
R2:

```
Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
O       10.1.3.0 [110/11] via 10.1.2.1, 00:01:24, FastEthernet1/0
C       10.1.2.0 is directly connected, FastEthernet1/0
C       10.2.3.0 is directly connected, FastEthernet0/0
O    192.168.1.0/24 [110/11] via 10.1.2.1, 00:01:24, FastEthernet1/0
C    192.168.2.0/24 is directly connected, FastEthernet0/1
O    192.168.3.0/24 [110/20] via 10.2.3.2, 00:01:24, FastEthernet0/0
```

R3:

```
Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
C       10.1.3.0 is directly connected, FastEthernet0/0
O       10.1.2.0 [110/2] via 10.2.3.1, 00:01:43, FastEthernet1/0
C       10.2.3.0 is directly connected, FastEthernet1/0
O    192.168.1.0/24 [110/12] via 10.2.3.1, 00:01:43, FastEthernet1/0
O    192.168.2.0/24 [110/11] via 10.2.3.1, 00:01:43, FastEthernet1/0
C    192.168.3.0/24 is directly connected, FastEthernet0/1
```

7.



R1:

interface Serial0/0

ip address 172.16.12.1 255.255.255.0

clock rate 2000000

R2:

interface Serial0/0

ip address 172.16.12.2 255.255.255.0

clock rate 2000000

!

interface Serial0/1

ip address 172.16.23.2 255.255.255.0

clock rate 2000000

R3:

interface Serial0/1

ip address 172.16.23.3 255.255.255.0

clock rate 2000000

8.

R1:

interface Loopback0

ip address 172.16.1.1 255.255.255.0

!

interface Loopback48

ip address 192.168.48.1 255.255.255.0

!

interface Loopback49

ip address 192.168.49.1 255.255.255.0

!

interface Loopback50

ip address 192.168.50.1 255.255.255.0

!

interface Loopback51

ip address 192.168.51.1 255.255.255.0

!

interface Loopback70

ip address 192.168.70.1 255.255.255.0

R2:

interface Loopback0

ip address 172.16.2.1 255.255.255.0

R3:

interface Loopback0

ip address 172.16.3.1 255.255.255.0

!

interface Loopback20

ip address 192.168.20.1 255.255.255.0

!

interface Loopback25

ip address 192.168.25.1 255.255.255.0

!

interface Loopback30

ip address 192.168.30.1 255.255.255.0

!

interface Loopback35

ip address 192.168.35.1 255.255.255.0

!

interface Loopback40

ip address 192.168.40.1 255.255.255.0

R1:

R1(config)#int s0/0

R1(config-if)#ip address 172.16.12.1 255.255.255.0

R1(config-if)#no fai

R1(config-if)#no fair-queue

R1(config-if)#clock

R1(config-if)#clock rat

R1(config-if)#clock rate 64000

R1(config-if)#no shut

R2:

R2(config)#int s0/0

R2(config-if)#ip address 172.16.12.2 255.255.255.0

R2(config-if)#no fair-q

R2(config-if)#no fair-queue

R2(config-if)#clock

R2(config-if)#clock rate 64000

R2(config-if)#no shut

R2(config-if)#int s0/1

R2(config-if)#ip address 172.16.23.2 255.255.255.0

R2(config-if)#no fair-queue

R2(config-if)#clock rate 64000

R2(config-if)#no shut

R3:

R3(config)#int s0/1

R3(config-if)#ip address 172.16.23.3 255.255.255.0

R3(config-if)#no fair-queue

R3(config-if)#clock rate 64000

R3(config-if)#no shut

9.

R1:

```
!
router rip
 version 2
 network 172.16.0.0
 network 192.168.48.0
 network 192.168.49.0
 network 192.168.50.0
 network 192.168.51.0
 network 192.168.70.0
 no auto-summary
```

R2:

```
!
router rip
 version 2
 network 172.16.0.0
 no auto-summary
!
```

R3:

```
!
router rip
 version 2
 network 172.16.0.0
 network 192.168.20.0
 network 192.168.25.0
 network 192.168.30.0
 network 192.168.35.0
 network 192.168.40.0
 no auto-summary
```

10.

R1:

```
R    192.168.30.0/24 [120/2] via 172.16.12.2, 00:00:28, Serial0/0
R    192.168.25.0/24 [120/2] via 172.16.12.2, 00:00:28, Serial0/0
R    192.168.40.0/24 [120/2] via 172.16.12.2, 00:00:28, Serial0/0
     172.16.0.0/24 is subnetted, 5 subnets
R       172.16.23.0 [120/1] via 172.16.12.2, 00:00:29, Serial0/0
C       172.16.12.0 is directly connected, Serial0/0
C       172.16.1.0 is directly connected, Loopback0
R       172.16.2.0 [120/1] via 172.16.12.2, 00:00:29, Serial0/0
R       172.16.3.0 [120/2] via 172.16.12.2, 00:00:01, Serial0/0
R    192.168.20.0/24 [120/2] via 172.16.12.2, 00:00:01, Serial0/0
C    192.168.51.0/24 is directly connected, Loopback51
C    192.168.50.0/24 is directly connected, Loopback50
R    192.168.35.0/24 [120/2] via 172.16.12.2, 00:00:01, Serial0/0
C    192.168.49.0/24 is directly connected, Loopback49
C    192.168.70.0/24 is directly connected, Loopback70
C    192.168.48.0/24 is directly connected, Loopback48
```

R2:

```
R    192.168.30.0/24 [120/1] via 172.16.23.3, 00:00:20, Serial0/1
R    192.168.25.0/24 [120/1] via 172.16.23.3, 00:00:20, Serial0/1
R    192.168.40.0/24 [120/1] via 172.16.23.3, 00:00:20, Serial0/1
     172.16.0.0/24 is subnetted, 5 subnets
C       172.16.23.0 is directly connected, Serial0/1
C       172.16.12.0 is directly connected, Serial0/0
R       172.16.1.0 [120/1] via 172.16.12.1, 00:00:10, Serial0/0
C       172.16.2.0 is directly connected, Loopback0
R       172.16.3.0 [120/1] via 172.16.23.3, 00:00:22, Serial0/1
R    192.168.20.0/24 [120/1] via 172.16.23.3, 00:00:22, Serial0/1
R    192.168.51.0/24 [120/1] via 172.16.12.1, 00:00:13, Serial0/0
R    192.168.50.0/24 [120/1] via 172.16.12.1, 00:00:13, Serial0/0
R    192.168.35.0/24 [120/1] via 172.16.23.3, 00:00:22, Serial0/1
R    192.168.49.0/24 [120/1] via 172.16.12.1, 00:00:14, Serial0/0
R    192.168.70.0/24 [120/1] via 172.16.12.1, 00:00:14, Serial0/0
R    192.168.48.0/24 [120/1] via 172.16.12.1, 00:00:14, Serial0/0
```

R3:

```
C     192.168.30.0/24 is directly connected, Loopback30
C     192.168.25.0/24 is directly connected, Loopback25
C     192.168.40.0/24 is directly connected, Loopback40
      172.16.0.0/24 is subnetted, 5 subnets
C        172.16.23.0 is directly connected, Serial0/1
R        172.16.12.0 [120/1] via 172.16.23.2, 00:00:11, Serial0/1
R        172.16.1.0 [120/2] via 172.16.23.2, 00:00:11, Serial0/1
R        172.16.2.0 [120/1] via 172.16.23.2, 00:00:11, Serial0/1
C        172.16.3.0 is directly connected, Loopback0
C     192.168.20.0/24 is directly connected, Loopback20
R     192.168.51.0/24 [120/2] via 172.16.23.2, 00:00:14, Serial0/1
R     192.168.50.0/24 [120/2] via 172.16.23.2, 00:00:14, Serial0/1
C     192.168.35.0/24 is directly connected, Loopback35
R     192.168.49.0/24 [120/2] via 172.16.23.2, 00:00:15, Serial0/1
R     192.168.70.0/24 [120/2] via 172.16.23.2, 00:00:15, Serial0/1
R     192.168.48.0/24 [120/2] via 172.16.23.2, 00:00:15, Serial0/1
```

As tabelas de encaminhamento contêm, para além das redes a que os routers estão diretamente conectados, também as redes anunciadas pelo RIP.

11.

R1:

```
R1#sh ip rip database
172.16.0.0/16    auto-summary
172.16.1.0/24    directly connected, Loopback0
172.16.2.0/24
    [1] via 172.16.12.2, 00:00:06, Serial0/0
172.16.3.0/24
    [2] via 172.16.12.2, 00:00:06, Serial0/0
172.16.12.0/24    directly connected, Serial0/0
172.16.23.0/24
    [1] via 172.16.12.2, 00:00:06, Serial0/0
192.168.20.0/24    auto-summary
192.168.20.0/24
    [2] via 172.16.12.2, 00:00:06, Serial0/0
192.168.25.0/24    auto-summary
192.168.25.0/24
    [2] via 172.16.12.2, 00:00:06, Serial0/0
192.168.30.0/24    auto-summary
192.168.30.0/24
    [2] via 172.16.12.2, 00:00:06, Serial0/0
192.168.35.0/24    auto-summary
192.168.35.0/24
    [2] via 172.16.12.2, 00:00:06, Serial0/0
192.168.40.0/24    auto-summary
192.168.40.0/24
    [2] via 172.16.12.2, 00:00:08, Serial0/0
192.168.48.0/24    auto-summary
192.168.48.0/24    directly connected, Loopback48
192.168.49.0/24    auto-summary
192.168.49.0/24    directly connected, Loopback49
192.168.50.0/24    auto-summary
192.168.50.0/24    directly connected, Loopback50
192.168.51.0/24    auto-summary
192.168.51.0/24    directly connected, Loopback51
192.168.70.0/24    auto-summary
192.168.70.0/24    directly connected, Loopback70
```

As redes anunciadas pelo router RIP são as redes 172.16.2.0, 172.16.3.0, 172.16.23.0, 192.168.20.0, 192.168.25.0, 192.168.30.0, 192.16.35.0 e 192.168.40.0

R2:

```
R2#sh ip rip database
172.16.0.0/16    auto-summary
172.16.1.0/24
    [1] via 172.16.12.1, 00:00:16, Serial0/0
172.16.2.0/24    directly connected, Loopback0
172.16.3.0/24
    [1] via 172.16.23.3, 00:00:26, Serial0/1
172.16.12.0/24    directly connected, Serial0/0
172.16.23.0/24    directly connected, Serial0/1
192.168.20.0/24    auto-summary
192.168.20.0/24
    [1] via 172.16.23.3, 00:00:26, Serial0/1
192.168.25.0/24    auto-summary
192.168.25.0/24
    [1] via 172.16.23.3, 00:00:26, Serial0/1
192.168.30.0/24    auto-summary
192.168.30.0/24
    [1] via 172.16.23.3, 00:00:26, Serial0/1
192.168.35.0/24    auto-summary
192.168.35.0/24
    [1] via 172.16.23.3, 00:00:26, Serial0/1
192.168.40.0/24    auto-summary
192.168.40.0/24
    [1] via 172.16.23.3, 00:00:26, Serial0/1
192.168.48.0/24    auto-summary
192.168.48.0/24
    [1] via 172.16.12.1, 00:00:18, Serial0/0
192.168.49.0/24    auto-summary
192.168.49.0/24
    [1] via 172.16.12.1, 00:00:18, Serial0/0
192.168.50.0/24    auto-summary
192.168.50.0/24
    [1] via 172.16.12.1, 00:00:18, Serial0/0
192.168.51.0/24    auto-summary
192.168.51.0/24
    [1] via 172.16.12.1, 00:00:18, Serial0/0
192.168.70.0/24    auto-summary
192.168.70.0/24
    [1] via 172.16.12.1, 00:00:18, Serial0/0
```

As redes anunciadas pelo router RIP são as redes 172.16.1.0, 172.16.3.0, 192.168.20.0, 192.168.25.0, 192.168.30.0, 192.168.35.0, 192.168.40.0, 192.168.48.0, 192.168.49.0, 192.168.50.0, 192.168.51.0 e 192.168.70.0

R3:

```
R3#sh ip rip database
172.16.0.0/16    auto-summary
172.16.1.0/24
    [2] via 172.16.23.2, 00:00:20, Serial0/1
172.16.2.0/24
    [1] via 172.16.23.2, 00:00:20, Serial0/1
172.16.3.0/24    directly connected, Loopback0
172.16.12.0/24
    [1] via 172.16.23.2, 00:00:20, Serial0/1
172.16.23.0/24    directly connected, Serial0/1
192.168.20.0/24    auto-summary
192.168.20.0/24    directly connected, Loopback2
192.168.25.0/24    auto-summary
192.168.25.0/24    directly connected, Loopback2
192.168.30.0/24    auto-summary
192.168.30.0/24    directly connected, Loopback3
192.168.35.0/24    auto-summary
192.168.35.0/24    directly connected, Loopback3
192.168.40.0/24    auto-summary
192.168.40.0/24    directly connected, Loopback4
192.168.48.0/24    auto-summary
192.168.48.0/24
    [2] via 172.16.23.2, 00:00:20, Serial0/1
192.168.49.0/24    auto-summary
192.168.49.0/24
    [2] via 172.16.23.2, 00:00:22, Serial0/1
192.168.50.0/24    auto-summary
192.168.50.0/24
    [2] via 172.16.23.2, 00:00:22, Serial0/1
192.168.51.0/24    auto-summary
192.168.51.0/24
    [2] via 172.16.23.2, 00:00:22, Serial0/1
192.168.70.0/24    auto-summary
192.168.70.0/24
    [2] via 172.16.23.2, 00:00:22, Serial0/1
```

As redes anunciadas pelo router RIP são as redes 172.16.1.0, 172.16.2.0, 172.16.12.0, 192.168.48.0, 192.168.49.0, 192.168.50.0, 192.168.51.0 e 192.168.70.0

13.

```
R2#sh ip protocols
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 9 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip
  Default version control: send version 2, receive version 2
    Interface             Send  Recv  Triggered RIP  Key-chain
    Serial0/0             2     2
    Loopback0             2     2
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    172.16.0.0
  Passive Interface(s):
    Serial0/1
  Routing Information Sources:
    Gateway         Distance      Last Update
    172.16.23.3          120      00:00:12
    172.16.12.1          120      00:00:07
  Distance: (default is 120)
```

14.

R2#config t

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#router rip

R2(config-router)#passive-interface s0/1

```
R2#sh ip protocols
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 9 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip
  Default version control: send version 2, receive version 2
    Interface             Send  Recv  Triggered RIP  Key-chain
    Serial0/0             2     2
    Loopback0             2     2
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    172.16.0.0
  Passive Interface(s):
    Serial0/1
  Routing Information Sources:
    Gateway         Distance      Last Update
    172.16.23.3          120      00:00:12
    172.16.12.1          120      00:00:07
  Distance: (default is 120)
```

Após desativar a interface s0/1, ele já não aparece listada para RIP

16. Tornar uma interface em RIP passiva apenas desabilita o envio de atualizações por meio de RIP; não afeta as interfaces recebidas por meio dele. Quais são alguns dos motivos pelos quais você deseja desativar o RIP de envio de atualizações para uma interface específica? Um dos motivos para tornar uma interface em RIP passiva é evitar tráfego desnecessário na rede enviado para uma interface que não está a ser usada em RIP. Outro dos motivos, é por questões de segurança pois um atacante poderia esta nessa porta e iria receber as atualizações do RIP conseguindo ver as rotas e redes todas

17. Colocar uma interface RIPv2 em modo passivo evita que o router envie pacotes RIP multicast por uma interface que não tenha vizinhos. O RIPv2 envia anúncios pelas interfaces de loopback? Sim, apesar das interfaces loopback não terem nenhuma ligação física, o router trata as interfaces de loopback da mesma forma que trata as interfaces físícas

18. Executando o RIPv2, deve-se implementar interfaces passivascomo uma prática comum para economizar os ciclos do processadorda CPU e a largura de banda em interfaces que não têmvizinhosRIPv2 multicast? Sim

19.

R2:

router ospf 1

log-adjacency-changes

network 172.16.23.0 0.0.0.255 area 0

R3:

router ospf 1

log-adjacency-changes

network 172.16.0.0 0.0.255.255 area 0

network 192.168.0.0 0.0.255.255 area 0

20.

R2:

```
R2#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.40.1      0   FULL/  -        00:00:37    172.16.23.3     Serial0/1
```

```
R2#sh ip route ospf
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.30.1/32 [110/65] via 172.16.23.3, 00:03:34, Serial0/1
     192.168.25.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.25.1/32 [110/65] via 172.16.23.3, 00:03:34, Serial0/1
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.40.1/32 [110/65] via 172.16.23.3, 00:03:34, Serial0/1
     172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
O       172.16.3.1/32 [110/65] via 172.16.23.3, 00:03:34, Serial0/1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.20.1/32 [110/65] via 172.16.23.3, 00:03:34, Serial0/1
     192.168.35.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.35.1/32 [110/65] via 172.16.23.3, 00:03:34, Serial0/1
```

R3:

```
R3#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
172.16.2.1        0   FULL/  -        00:00:36    172.16.23.2     Serial0/1
```

```
R3#sh ip route ospf

```

21.

```
R2#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
R       192.168.30.0/24 [120/1] via 172.16.23.3, 00:00:06, Serial0/1
O       192.168.30.1/32 [110/65] via 172.16.23.3, 00:15:03, Serial0/1
     192.168.25.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.25.1/32 [110/65] via 172.16.23.3, 00:15:03, Serial0/1
R       192.168.25.0/24 [120/1] via 172.16.23.3, 00:00:06, Serial0/1
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
R       192.168.40.0/24 [120/1] via 172.16.23.3, 00:00:07, Serial0/1
O       192.168.40.1/32 [110/65] via 172.16.23.3, 00:15:04, Serial0/1
     172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C       172.16.23.0/24 is directly connected, Serial0/1
C       172.16.12.0/24 is directly connected, Serial0/0
R       172.16.1.0/24 [120/1] via 172.16.12.1, 00:00:40, Serial0/0
O       172.16.3.1/32 [110/65] via 172.16.23.3, 00:15:06, Serial0/1
C       172.16.2.0/24 is directly connected, Loopback0
R       172.16.3.0/24 [120/1] via 172.16.23.3, 00:00:09, Serial0/1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
R       192.168.20.0/24 [120/1] via 172.16.23.3, 00:00:09, Serial0/1
O       192.168.20.1/32 [110/65] via 172.16.23.3, 00:15:06, Serial0/1
R    192.168.51.0/24 [120/1] via 172.16.12.1, 00:00:42, Serial0/0
R    192.168.50.0/24 [120/1] via 172.16.12.1, 00:00:42, Serial0/0
     192.168.35.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.35.1/32 [110/65] via 172.16.23.3, 00:15:06, Serial0/1
R       192.168.35.0/24 [120/1] via 172.16.23.3, 00:00:09, Serial0/1
R    192.168.49.0/24 [120/1] via 172.16.12.1, 00:00:42, Serial0/0
R    192.168.70.0/24 [120/1] via 172.16.12.1, 00:00:42, Serial0/0
R    192.168.48.0/24 [120/1] via 172.16.12.1, 00:00:42, Serial0/0
```

Não existe uma super-rede 192.168.0.0, pois o OSPF está a anunciar cada sub-rede individualmente com o endereço de loopback e máscara /32

23.

```
R3#conf t
Enter configuration commands, one per line.  En
R3(config)#router ospf 1
R3(config-router)#passi
R3(config-router)#passive-interface loopback 0
```

24.

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#passive-interface default
R3(config-router)#
*Mar  1 00:34:59.407: %OSPF-5-ADJCHG: Process 1, Nbr 172.16.2.1 on Serial0/1 from FULL to DOWN
, Neighbor Down: Interface down or detached
R3(config-router)#no passive-interface s0/1
R3(config-router)#
*Mar  1 00:35:14.259: %OSPF-5-ADJCHG: Process 1, Nbr 172.16.2.1 on Serial0/1 from LOADING to F
ULL, Loading Done
```

25.

```
  Passive Interface(s):
    FastEthernet0/0
    Serial0/0
    FastEthernet0/1
    Loopback0
    Loopback20
    Loopback25
  Passive Interface(s):
    Loopback30
    Loopback35
    Loopback40
    VoIP-Null0
```

Ao fazer show ip protocols, verifica-se que todas as interfaces estão em modo passivo à exceção da interface Serial 0/1

26.

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router rip
R2(config-router)#redistribute ospf 1 metric 4
R2(config-router)#end
```

27.

```
R2#sh ip protocols
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 22 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip, ospf 1
  Default version control: send version 2, receive version 2
    Interface             Send  Recv  Triggered RIP  Key-chain
    Serial0/0             2     2
    Loopback0             2     2
```

O R2 está a redistribuir as rotas OSPF

28.

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#redistribute rip
% Only classful networks will be redistributed
```

29.

```
R3#sh ip route ospf
O E2 192.168.51.0/24 [110/20] via 172.16.23.2, 00:01:59, Serial0/1
O E2 192.168.50.0/24 [110/20] via 172.16.23.2, 00:01:59, Serial0/1
O E2 192.168.49.0/24 [110/20] via 172.16.23.2, 00:01:59, Serial0/1
O E2 192.168.70.0/24 [110/20] via 172.16.23.2, 00:01:59, Serial0/1
O E2 192.168.48.0/24 [110/20] via 172.16.23.2, 00:01:59, Serial0/1
```

Ao contrário do que acontece no guião, onde a única rota OSPF externa é a da rede 192.168.70.0/24, neste caso, para além desta rota também inclui a rotas para as redes loopback do R1

30.

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#redistribute rip subnets
R2(config-router)#end
R2#
```

```
R3#sh ip route ospf
     172.16.0.0/24 is subnetted, 5 subnets
O E2    172.16.12.0 [110/20] via 172.16.23.2, 00:00:35, Serial0/1
O E2    172.16.1.0 [110/20] via 172.16.23.2, 00:00:35, Serial0/1
O E2    172.16.2.0 [110/20] via 172.16.23.2, 00:00:35, Serial0/1
O E2 192.168.51.0/24 [110/20] via 172.16.23.2, 00:11:12, Serial0/1
O E2 192.168.50.0/24 [110/20] via 172.16.23.2, 00:11:12, Serial0/1
O E2 192.168.49.0/24 [110/20] via 172.16.23.2, 00:11:12, Serial0/1
O E2 192.168.70.0/24 [110/20] via 172.16.23.2, 00:11:12, Serial0/1
O E2 192.168.48.0/24 [110/20] via 172.16.23.2, 00:11:12, Serial0/1
```

Á semelhança da situação anterior, são apresentadas as rotas para as redes loopback do R1, mas agora também são apresentadas as rotas para as redes 172.16.12.0, 172.16.1.0 e 172.16.2.0

31.

Ping do R1 para o Loopback 0 do R3:

```
R1#ping 172.16.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/31/36 ms
```

Ping do R3 para o Loopback 0 do R1:

```
R3#ping 172.16.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
```

32.

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#default-metric 10000
```

33.

```
R3#sh ip route ospf
     172.16.0.0/24 is subnetted, 5 subnets
O E2    172.16.12.0 [110/10000] via 172.16.23.2, 00:00:31, Serial0/1
O E2    172.16.1.0 [110/10000] via 172.16.23.2, 00:00:31, Serial0/1
O E2    172.16.2.0 [110/10000] via 172.16.23.2, 00:00:31, Serial0/1
O E2 192.168.51.0/24 [110/10000] via 172.16.23.2, 00:00:31, Serial0/1
O E2 192.168.50.0/24 [110/10000] via 172.16.23.2, 00:00:31, Serial0/1
O E2 192.168.49.0/24 [110/10000] via 172.16.23.2, 00:00:31, Serial0/1
O E2 192.168.70.0/24 [110/10000] via 172.16.23.2, 00:00:31, Serial0/1
O E2 192.168.48.0/24 [110/10000] via 172.16.23.2, 00:00:31, Serial0/1
```

Após mudar o custo default das rotas para 10000, verifica-se que agora todas as rotas OSPF tem, por padrão, custo 10000

34.

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#redistribute rip sub metric-type 1
```

35.

```
R3#sh ip route ospf
     172.16.0.0/24 is subnetted, 5 subnets
O E1    172.16.12.0 [110/10064] via 172.16.23.2, 00:00:36, Serial0/1
O E1    172.16.1.0 [110/10064] via 172.16.23.2, 00:00:36, Serial0/1
O E1    172.16.2.0 [110/10064] via 172.16.23.2, 00:00:36, Serial0/1
O E1 192.168.51.0/24 [110/10064] via 172.16.23.2, 00:00:36, Serial0/1
O E1 192.168.50.0/24 [110/10064] via 172.16.23.2, 00:00:36, Serial0/1
O E1 192.168.49.0/24 [110/10064] via 172.16.23.2, 00:00:36, Serial0/1
O E1 192.168.70.0/24 [110/10064] via 172.16.23.2, 00:00:36, Serial0/1
O E1 192.168.48.0/24 [110/10064] via 172.16.23.2, 00:00:36, Serial0/1
```

Verificamos que o tipo mudou de 2 para 1